

LECTURE 1:

INTRODUCTION

COMP2013J: Databases and Information Systems

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A little bit about me

- **Lecturer:** Dr. Ruihai Dong (ruihai.dong@ucd.ie)
- PhD Computer Science, University College Dublin (UCD)
Mining Opinionated Recommendations for and from Customer Reviews
- 2014 - 2016: Postdoctoral Researcher
Insight Centre for Data Analytics, UCD, Ireland
- 2016 - Present: Assistant Professor, Associate Professor
School of Computer Science, UCD, Ireland
- 2019 - Present: Funded Investigator
Insight Centre for Data Analytics II, UCD, Ireland
- 2019 - Present: Named Supervisor
SFI Centre for Research Training in Machine Learning, Ireland
- **Google Scholar:** <https://scholar.google.com/citations?user=icGRQ68AAAAJ&hl=en>

Research Interests

- **Recommender Systems:** A few interesting and challenged subtopics, such as deep content-based recommendations, transfer learning-based domain-independent recommender systems, graph-based recommendations, etc.
- **AI-first Finance:** Aims to leverage cutting-edge artificial intelligence and natural language processing technologies to automatically analyze structured and unstructured data and build novel interpretable financial forecasting models.

Head TA



Jie Chen

PhD student (Insight Center)

`jie.chen1@ucdconnect.ie`

Visualisation of High-dimensional
Data in Deep Learning

<https://www.insight-centre.org/our-team/jie-chen-phd-student/>

Module Format

- Lectures will run from week 1 to week 15 (inclusive).
- Labs will run from week 3 to week 15 (inclusive).
- If you have any questions about the module, please ask me or a teaching assistant (TA).
- After a lecture, during labs or by email:
 - My email address is ruihai.dong@ucd.ie
 - Jie Chen's email address is: jie.chen1@ucdconnect.ie
- Assessment
 - 30% Assignment
 - 70% Final Exam

Practical Work and Assignments

- Weekly worksheets are **important**.
- Major Assignment (**30%**) is **Group work** (group assignment).
- 5 members each team.
- Plagiarism is a **very serious** academic offence.



Academic Integrity & UCD Computer Science

- **Academic misconduct is "any attempt by someone to seek unfair advantage in relation to [an] academic activity or which facilitates others to gain an unfair advantage, or to profit from the sharing or selling of your own or others' work without permission" [[UCD Academic Integrity Policy](#)]**
 - **Examples of academic misconduct include but are not limited to:**
 - plagiarism, which can take many forms e.g.
 - copying some/all of the work of another student and submitting it as your own work
 - copying some/all of an assignment from the Internet/book/etc without referencing it
 - sharing individual work with another student (by e-mail, FB messenger, WhatsApp, ...)
 - **submission of AI-generated content without explicit permission and attribution**
 - students collaborating at too detailed a level: consulting each other after implementing a line/block/segment of code, sharing results, then individually submitting the resulting work
 - collusion (undisclosed collaboration not permitted by the lecturer)
 - publishing, uploading or sharing submitted work without the lecturer's permission
 - copying or cheating in written examinations
 - advertising or publicising cheating services

Student Conduct and Academic Integrity School of CS Academic Integrity Policy

- CS staff/demonstrators are **proactive** on academic misconduct → **don't challenge them!**
- Student who **enables** academic misconduct is **equally responsible** for it

Module Content

- Reading List
- Fundamentals of Database Systems (7th Edition)
 - ISBN-10: 0133970779 | ISBN-13: 978-0133970777
- Database Systems: A Practical Approach to Design, Implementation, and Management (6th Edition)
 - ISBN-10: 0132943263 | ISBN-13: 978-0132943260
- All materials will be covered in lecture notes
- Practical work will use MySQL Relational DataBase Management System (RDBMS)
 - Free, cross platform, open source database management system

Topics

- Introduction to Databases
- Database Models
- Relational Database Model
- Structured Query Language (1974, 1981)
- Programmatic DB Use – Build you own first Information System
- Database Design (Entity-Relationship Model)
- Database Normalisation
- Object Relational Mapping

DATABASE AND DBMS

Why Study Databases?

- A huge amount of information being stored.
- The College, Medical records, Employers, Companies, Government Agencies, etc.
- Managing that data is a **really big** task
- Data Base Management Systems (DBMS)
- Storing is easy, **managing** is the issue

What is a Database?

- Initial Definition: A database is a collection of related data
- This is a collection of related data with an implicit meaning and hence is a database
- For example, consider the names, telephone numbers, and addresses of the people you know

What is a Database?

- You may have recorded this data in an indexed address book (The contacts in your phone)



More Specific Properties of a Database

- A database represents some aspect of the real world
 - Changes to the real world are reflected in the database



David	1,000\$
Ruihai	300\$

More Specific Properties of a Database

- A database is a logically connected collection of data with some meaning
 - A random assortment of data cannot correctly be called a database

Place	County	Phone code	Approx. population
Basingstoke	Hampshire	01256	82913
Brighton	East Sussex	01273	155919
Carlisle	Cumbria	01228	103700
Huddersfield	Yorkshire	01484	146234
Luton	Bedfordshire	01582	203800
Nottingham	Nottinghamshire	0115	292400
Rhyl	Clwyd	01745	24889
Woking	Surrey	01483	62796

Zhang San 010-2895331

Ha Ha :D

Math 85%

An example of  database

- A database is designed, built, and populated with data for a specific purpose
 - It has an intended group of users and some applications in which these users are interested

Database Management Systems

- A database management system (DBMS) is a collection of programs that enables users to create and maintain a database
- The DBMS is a general-purpose software system that allows the processes of **defining**, **constructing**, **manipulating**, and **sharing** databases among various users and applications

DBMS Models

- Nowadays almost all databases are **Relational**
- This is a specific model and this module we will concentrate on this model
- Other models that have been used are
 - Hierarchical
 - Network
- Other models have been suggested in recent years
 - Object-Oriented

DBMS MODELS

Managing Data

- Businesses have always maintained data
- For centuries this was done on paper

“The palest ink is better than the best memory.” (Chinese proverb)

好记忆不如烂笔头

- Huge files that had to be manually searched in order to find information

Hard to Retrieve



Managing Data

- The advent of computers allowed electronic storage
- First attempts stored electronic versions of documents
- The first major problem was how to store financial information

Spreadsheets



Spreadsheets

- Spreadsheets have been used by accountants for centuries.
- Very early electronic versions of spreadsheets were developed for mainframe computers in the 1960s.
- Modern computerised spreadsheets began with Daniel Bricklen and Bob Frankston.
- They developed VisiCalc in 1978, which became the basis for all electronic spreadsheets since.

VisiCalc

A1	(C) 1979,1981 Software Arts, Inc. UC-176Y2-IBM-TEST							C
	#00000000							43
	A	B	C	D	E	F	G	H
1		ITEM		NO.	UNIT		COST	
2		---		---	---		---	
3								
4	MUCK	RAKE		43	12.95		556.85	
5	BUZZ	CUT		150	6.75		1012.50	
6								
7	TOE	TONER		250	49.95	12487	31487.50	
8	EYE	SNUFF		2	4.95		9.90	
9								
10								
11					SUBTOTAL	13155	55.50	
12					9.75% TAX		1282.66	
13								
14					TOTAL	14438	.16	
15								
16								
17								
18								
19								
20								
21								

Storing Data

- Storing data as simply electronic copies of existing documents is not sufficient.
- How do we search for information?
 - How many BSc II students do we have?
 - Is Ruihai Dong in the list?
 - What modules is he registered for?
- This type of query requires that we change how things are done.
- People then began to look at how to model data for electronic use.

File Based Database

Sales File

001-100-Staples
002-5000-Paper
003-1-Printer

Program



Purchases File

101-10000-Paper
102-200-Staples
...

Program



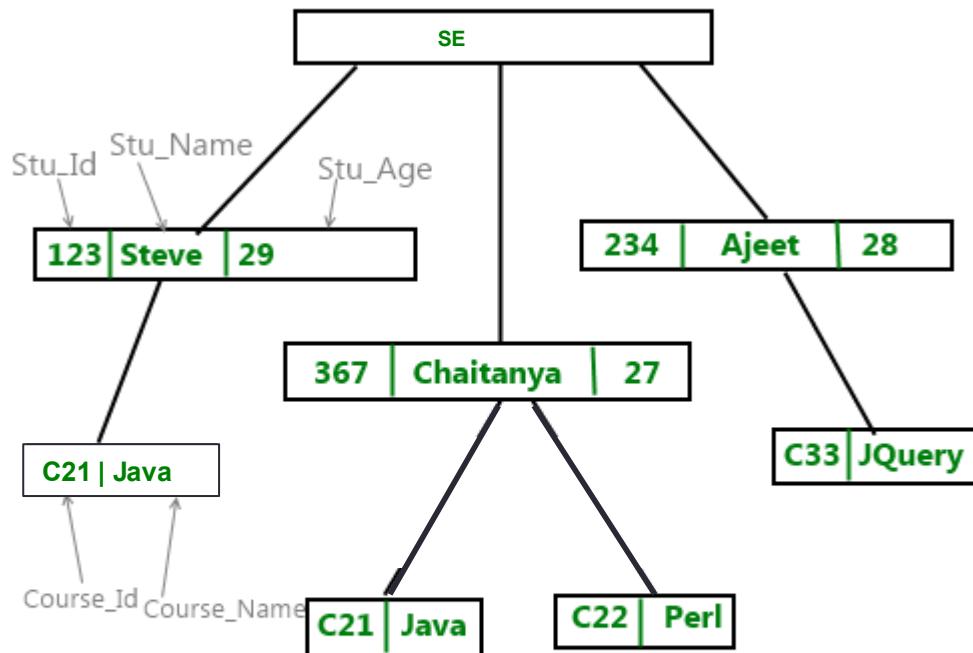
File Based Database Systems

- Advantages
 - More lightweight than a DBMS
 - Might be good enough for small data, for example a personal address book
- Disadvantages
 - No query language
 - No scalability
 - Hard to update schema and modify data
 - Recovery?

Hierarchical Database Model

- The oldest DBMS model is the hierarchical model (Information Management System from IBM)
- Developed in the 1960s to overcome the problems with file processing
- The model was not standardised
- It was based on a tree structure consisting of nodes, branches and roots
- It allowed for 1 to many relationships
- The best known implementation is IMS by IBM

Hierarchical Based Database



The layout is highly efficient for operations that “drill down,” so queries such as “how many courses does Chaitanya choose?” are easy to answer.

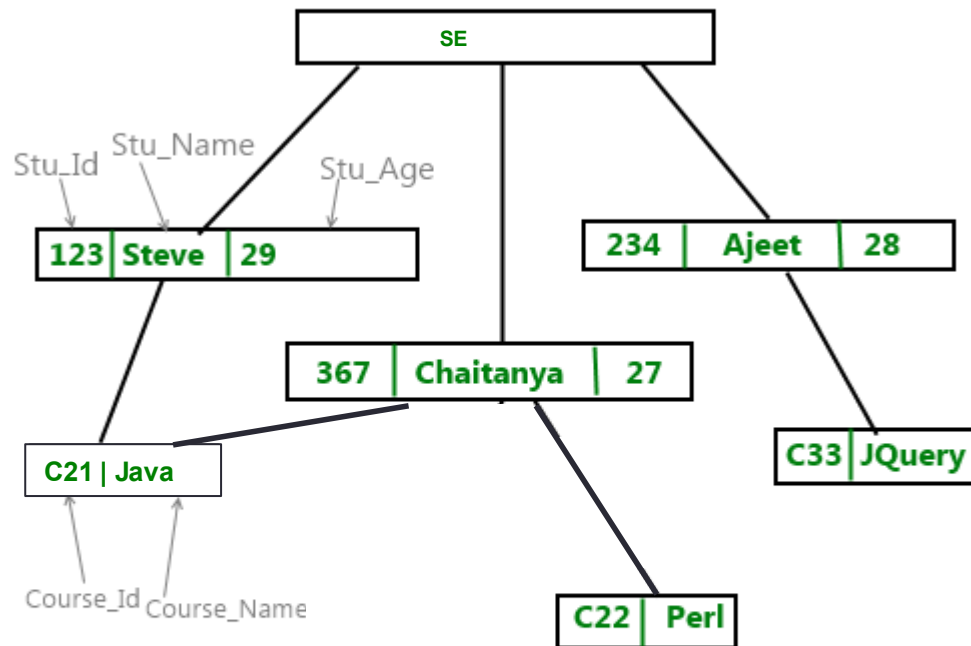
However, non-hierarchical queries are difficult to express and very costly to evaluate: queries such as “how many students do study Java?”

The hierarchical model is also prone to redundancy.

Network Database System

- The network model was developed by committee
- The Database Task Group (DBTG) of CODASYL (COncference on DAta SYstems Languages) developed the model so that it could be standardised
- CODASYL are the people who developed the COBOL programming language and much of the language of the network model looks like COBOL
- The model allows the implementation of many-to-many (M:M) relationships, which ultimately get translated into one-to-many (1:M).

Network Database System



The network model generalizes the hierarchical model to represent relationships as directed graphs rather than trees. Doing so improves efficiency significantly: an application can easily add connections between records in order to accelerate important queries.

This model also has some problems, the main problem being that you need to be a database expert to use this database successfully. It is very difficult for the general public to use.

Relational Database Model

- Formulated by Edgar Codd of IBM in 1970.
- Commercial RDBMS in 80s.
- “Codd's 12 Rules” (actually 13) that all RDBMSs should follow.
- Most widely used Model at present
 - Access, Oracle, MySQL, MariaDB, MS SQL Server, DB2, Sybase ASE, PostgreSQL etc.

RELATIONAL DATABASE MODEL

Relational Concepts

- Data is represented as a collection of **relations**
- Each relation is **table** of values
- Each table consists of **rows** and **columns**

Student

<u>StudentNo</u>	FirstName	LastName	YearofEntry	Major
1312345	Lina	Xu	2013	Finance
1318999	Wan	Wan	2013	Software Engineering
1218985	Ning	Cao	2012	Internet of Things

Relational Concepts

- Each **row** represents an **entity** or **record**
- Rows are unordered

<u>StudentNo</u>	FirstName	LastName	YearofEntry	Major
1312345	Lina	Xu	2013	Finance
1318999	Wan	Wan	2013	Software Engineering
1218985	Ning	Cao	2012	Internet of Things

Relational Concepts

- No duplicate rows are allowed
- Each relation has a **primary key**, the value of which uniquely identifies the **record/entity**

<u>StudentNo</u>	FirstName	LastName	YearofEntry	Major
1312345	Lina	Xu	2013	Finance
1318999	Wan	Wan	2013	Software Engineering
1218985	Ning	Cao	2012	Internet of Things

1218985	Ning	Ca 	2012	Internet of Things
1218986	Ning	Cao	2012	Internet of Things

Relational Concepts

- Each column represents an **attribute**
- Table name and column name are used to help interpret the values

<u>StudentNo</u>	StudentName	Major	YearofEntry
1312345	Xu Lina	Finance	2013
1318999	Jie Wan	Software Engineering	2013
1218985	Cao Ning	Internet of Things	2012

Database Terminology

- **Relation** is a mathematical term for a **table**
- **Row** is called a **Tuple**
- **Column** is called an **Attribute**
- **Domain** is used to describe the types of values that can appear in a column
- **Degree** is the number of attributes
- **Cardinality** – the number of tuples/rows in a relation
- **Atomic Value** – precisely one value at each row intersection
- **Null Value** – Missing, not known or irrelevant data (not the same as zero or blank)

Student Table/Relation

Relation
Name

Students

Primary
Key

Attributes

Degree is 5

<u>StudentNo</u>	FirstName	LastName	YearofEntry	Major
1312345	Lina	Xu	2013	Finance
1318999	Wan	Wan	2013	Software Engineering
1218985	Ning	Cao	2012	Internet of Things

Tuples

Cardinality
is 3

Domain of Major Attribute is =
{Finance, Software Engineering,
Internet of Things}

Advantages of Database Approach

- Data can be shared
- Redundancy can be reduced
- Integrity can be maintained
- Security can be enforced
- Conflicting requirements can be balanced
- Standards can be enforced